

Problem Set 8

Right-handed Neutrinos

Suppose a right-handed neutrino for each generation is added to Standard Model.

1) Show that the only new renormalizable term that can appear in the Lagrangian is

$$-\frac{1}{2}\bar{L}_m \not{D} L_m - \frac{1}{2}\bar{N}_m \not{D} N_m - \frac{1}{2}M_m \bar{N}_m N_m$$

Lepton doublet $\swarrow$

$$-(k_{mn} \bar{L}_m P_R N_n \tilde{\Phi} + \text{h.c.})$$

$\nwarrow$ Majorana spinor

whose right-handed part is the right-handed neutrino

2) Do any combinations of electron-number, muon-number & tau-number remain preserved?

3) If there's only one generation what's the mass matrix for the neutrino.

4) Express the Lepton-Higgs & Lepton-gauge boson interactions in terms of these mass eigenstates.

2. Adjoint Representation fermions

Suppose two Majorana fermions were added $\tilde{W} (1, 3, 0)$ with transf.

$\tilde{G} (8, 1, 0)$ properties

$$\delta P_L \tilde{W} = -\epsilon_{abc} \omega_2^b P_L \tilde{W}^c$$

$$D_\mu P_L \tilde{W}^a = (\partial_\mu \delta_{ac} + g_2 \epsilon_{abc} W_\mu^b) P_L \tilde{W}^c$$

and

$$\delta P_L \tilde{G}^x = -f_{\alpha\beta\gamma} \omega_3^\beta P_L \tilde{G}^\gamma$$

$$D_\mu P_L \tilde{G}^x = (\partial_\mu \delta_{\alpha\gamma} + g_3 f_{\alpha\beta\gamma} G_\mu^\beta) P_L \tilde{G}^\gamma$$

1) Show that the reality of G_{abc} & $f_{\alpha\beta\gamma}$ cause $P_R \tilde{W} \propto P_R \tilde{G}$ to have the same transf. as $P_L \tilde{W} \propto P_L \tilde{G}$

2) Show that these adjoint fermions do have $SU(3)_C \times SU_L(2) \times U(1)_Y$ invariant mass terms

$$-\frac{m_{\tilde{W}}}{2} \overline{\tilde{W}} \tilde{W} - \frac{m_{\tilde{G}}}{2} \overline{\tilde{G}} \tilde{G}$$

3) Are there new Yukawa interactions? what are they?